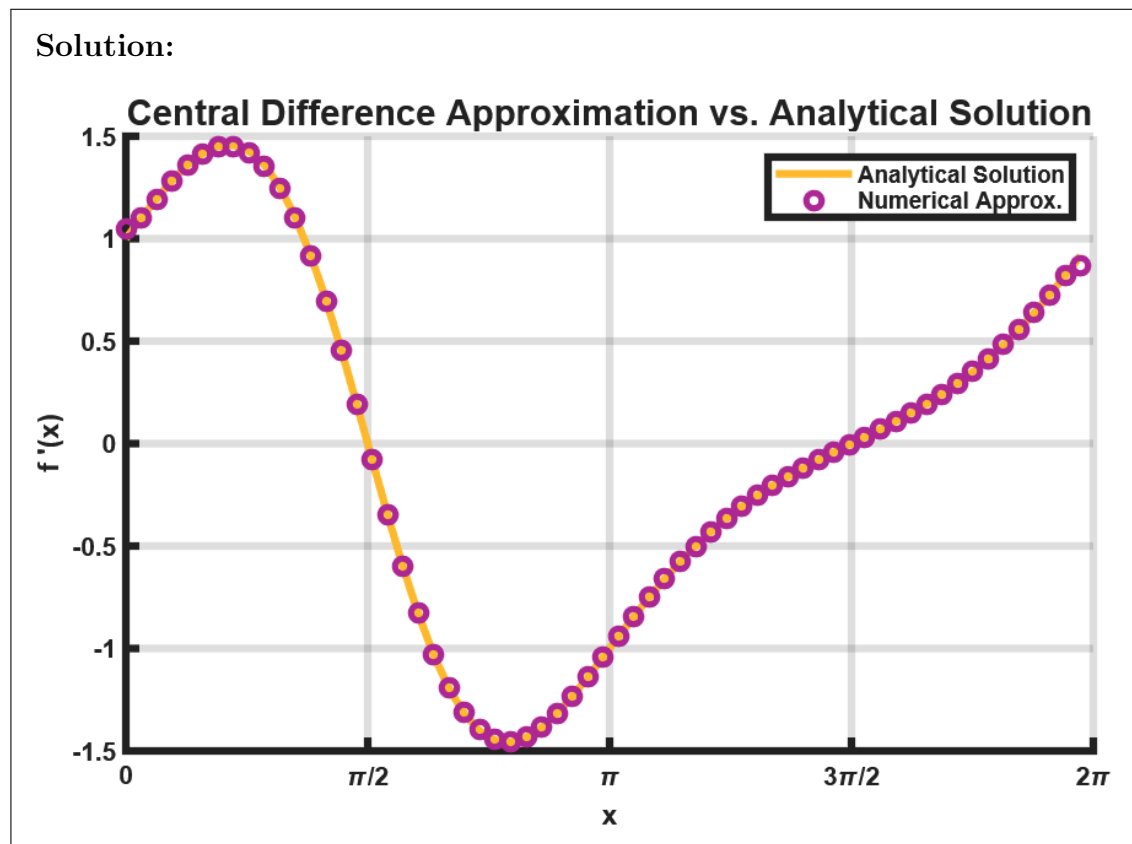


1. Write a matlab code that uses finite differences to compute the first derivative of

$$f(x) = \exp(\sin(x))$$

on the interval $0 \leq x \leq 2\pi$. Use second-order centered differences on internal grid points and first-order one-sided differences at endpoints.

- (a) Show a graph of the exact analytic solution $f'(x)$ and your numerical solution $f'_n(x)$.



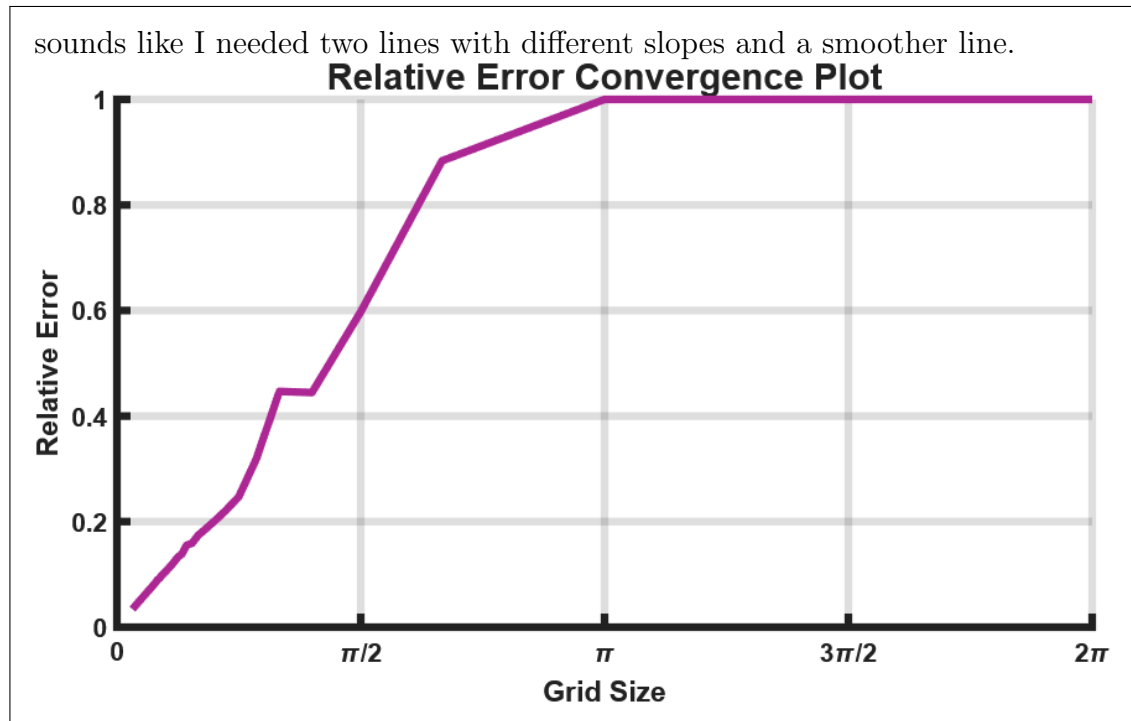
- (b) Show a relative error convergence plot by plotting the relative error on the y-axis and the grid size N on the x-axis. Measure relative error as

$$Err = \frac{\|f'(x) - f'_n(x)\|_p}{\|f'(x)\|_p}.$$

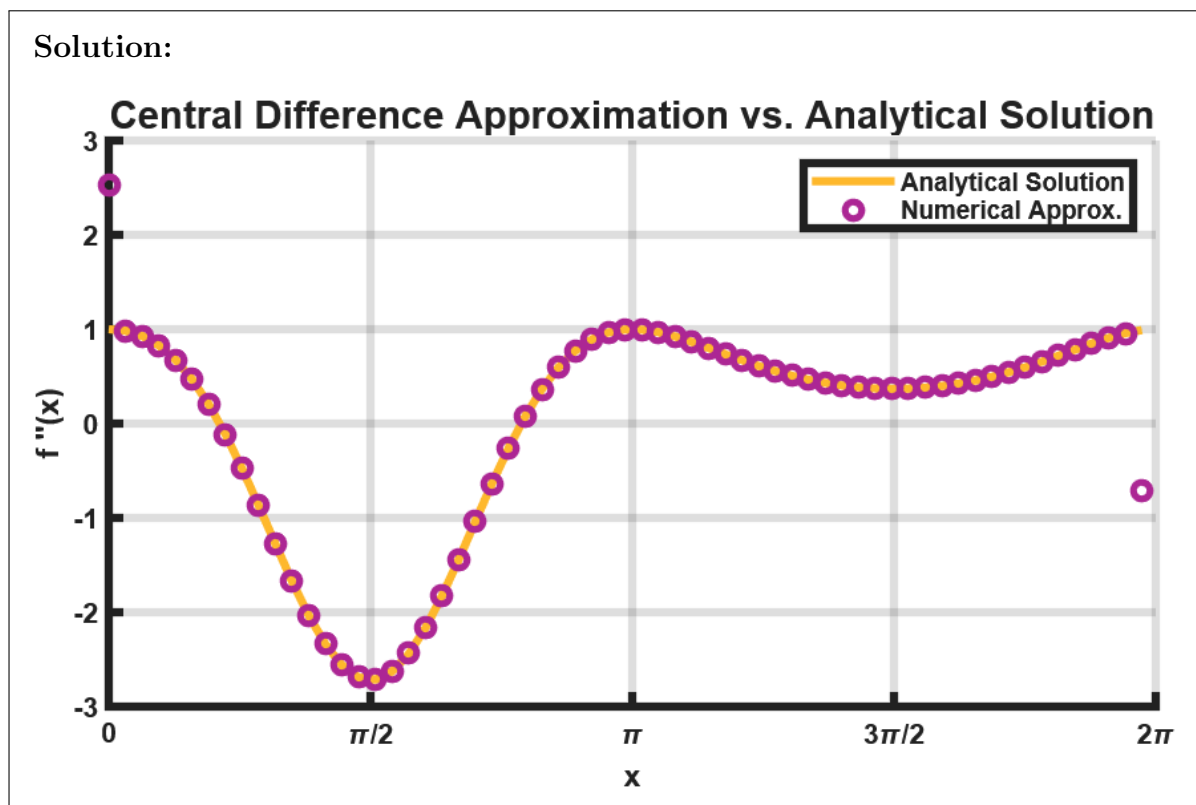
Plot the error using both the $p = \infty, 2$ norms in a log-log scale. The relative errors should be lines with slopes 1 and $3/2$ respectively, i.e. it scales with $\frac{1}{N}$, $\frac{1}{N^{3/2}}$. Extra Credit: The $3/2$ slope is strange, why is that happening?

Solution:

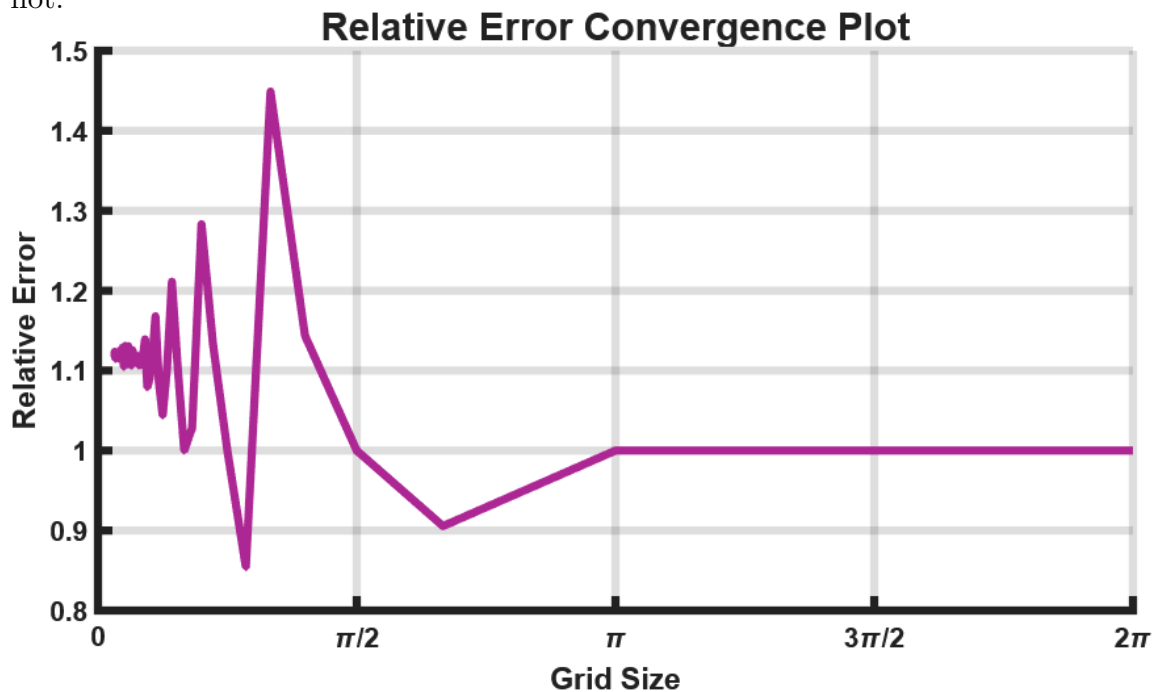
As a general note, I was unsure of what this problem was asking me to plot. It



2. Repeat problem 1 but compute the second derivative $f''(x)$. Assume periodicity and wrap around the grid instead of using one-sided finite differences.



As a note, I checked my endpoints many times and am unsure if they are correct or not.



Again, I am unsure of what the error convergence plot is supposed to be. I do have my code files included in my github repo, and hoping I can work to improve my work and resubmit.

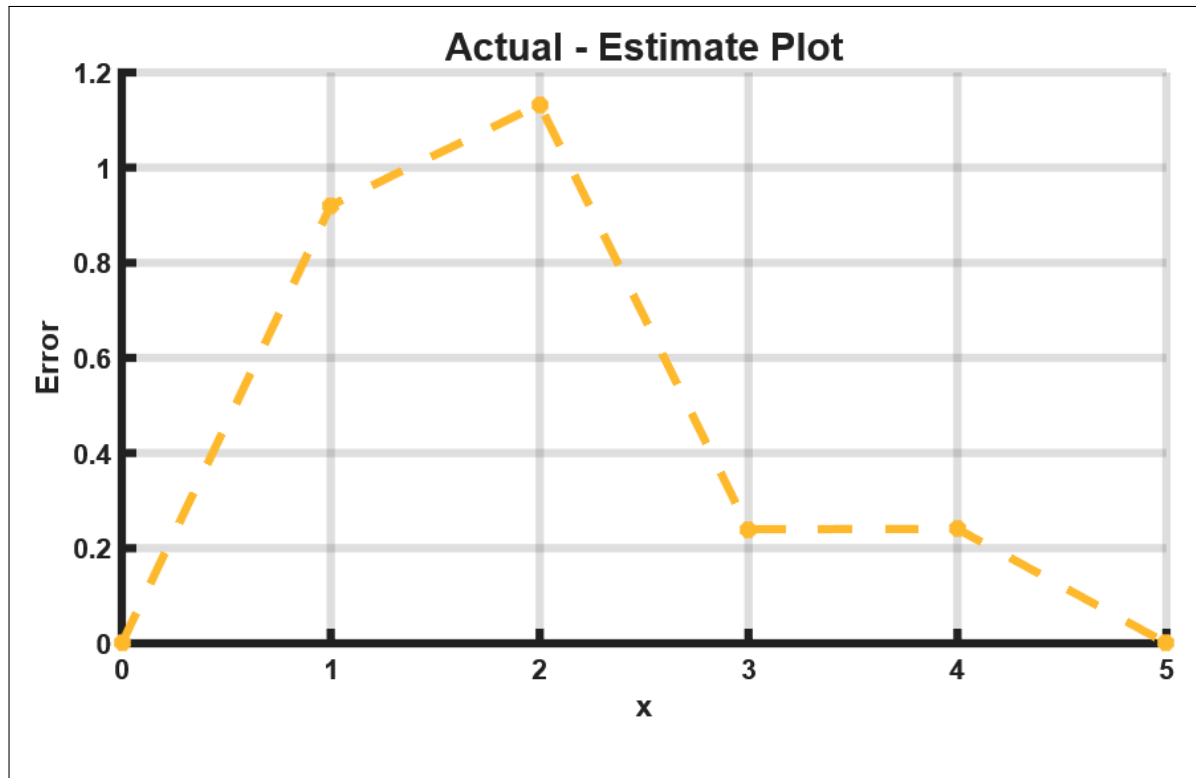
3. Consider the ODE

$$\frac{d^2u}{dx^2} + \sin(x) \frac{du}{dx} + u(x) = f(x)$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with the Dirichlet conditions. Use $u(x) = \sin(x)$ to test your solution by finding boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.

Solution:

This is not an error convergence plot, but I was unable to implement this correctly in Matlab in time. Here is a plot of the absolute error for our ODE where the spacing $h = 1$ on $0 \leq x \leq 5$.



4. Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5], y \in [0, 5]$. Use the test solution $u(x, y) = \sin(x) \cos(y)$.
- (a) Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second order convergence plot (similar to the first three problems).

Solution: I don't know where to start coding with this one, so I'm just going to conceptually explain what I understand with this problem. We have our boundaries, so we can choose i to be our x-axis on our grid and j to be our y-axis, with positive values being right and up respectively. To discretize our finite difference method, we let our function $f_{i,j} = \frac{U_{i,j+1} - 2U_{i,j} + U_{i,j-1}}{\Delta x^2} + \frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{\Delta y^2}$. To code, we choose i or j to be the inner loop ("fast index") and the other to be the outer loop ("slow index"). We calculate each $U_{i,j}$ and use the U matrix to solve for the vector of unknowns.

- (b) Modify your code to apply Neumann conditions on all boundaries. This won't work. Why not? - both in terms of the linear algebra and in terms of the actual PDE we want to solve. Suggest a solution to the issue.

Solution: The Neumann conditions are when we know the derivative's boundaries. Applying them without modifying the rest of our code would not work,

because the Neumann conditions has a different matrix and rhs vector format.